

Lecture 14: Number System Conversions (Part 1): Decimal $\leftrightarrow$ Binary

Learning Objectives

By the end of this lecture, students will be able to:

- Convert Decimal numbers to Binary.
 - Convert Binary numbers to Decimal.
 - Understand the logic behind each conversion.
 - Solve conversion problems step by step.
 - Avoid common mistakes in university exams.
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1. Introduction

Teaching Script

First 30-Second SEO Phrase

Naturally include:

"In today's Computer Organization and Assembly Language lecture, we'll learn Decimal to Binary and Binary to Decimal conversion using the repeated division method and positional weight method, with easy step-by-step examples for beginners."

In the previous lecture, we learned about Decimal, Binary, Octal, and Hexadecimal number systems.

Today, we're starting one of the most important practical topics in Computer Organization—**Number System Conversions**.

If you want to understand Assembly Language, memory addressing, machine instructions, and digital electronics, you must know how to convert numbers between different number systems.

In today's lecture, we'll focus only on **Decimal to Binary** and **Binary to Decimal** conversions. We'll solve several examples together using easy step-by-step methods.

Let's begin.

2. Why Do We Need Number Conversions?

Humans naturally work with **Decimal** numbers.

Example:

25

100

255

500

However, computers understand only **Binary**.

11001

1100100

11111111

111110100

Whenever we enter a decimal number, the computer internally converts it into binary before processing it.

3. Decimal → Binary Conversion

Method: Repeated Division by 2

Steps

1. Divide the decimal number by **2**.
 2. Write the **remainder**.
 3. Continue dividing the quotient by **2**.
 4. Stop when the quotient becomes **0**.
 5. Read the remainders **from bottom to top**.
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Example 1

Convert:

$13_{10} \rightarrow \text{Binary}$

Solution

Division	Quotient	Remainder
$13 \div 2$	6	1
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Read from **bottom to top**:

1101_2

Answer:

$13_{10} = 1101_2$

Example 2

Convert:

25_{10}

Division	Quotient	Remainder
$25 \div 2$	12	1
$12 \div 2$	6	0
$6 \div 2$	3	0
$3 \div 2$	1	1
$1 \div 2$	0	1

Bottom → Top:

11001_2

Example 3

Convert:

45_{10}

Division	Quotient	Remainder
$45 \div 2$	22	1
$22 \div 2$	11	0
$11 \div 2$	5	1
$5 \div 2$	2	1
$2 \div 2$	1	0
$1 \div 2$	0	1

Reading upward:

101101_2

4. Shortcut Method (Using Powers of 2)

Instead of repeated division, identify the powers of 2 that make up the decimal number.

Example:

Convert:

26_{10}

Largest power of 2 less than or equal to 26:

16

$$26 - 16 = 10$$

Next:

8

$$10 - 8 = 2$$

Next:

2

$$2 - 2 = 0$$

Place 1s and 0s:

16	8	4	2	1
1	1	0	1	0

Result:

11010₂

5. Binary → Decimal Conversion

Method: Positional Weights

Steps

1. Write the powers of 2 from right to left.
 2. Multiply each binary digit by its corresponding power.
 3. Add all the results.
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Example 1

Convert:

1101₂

Binary Digit	1	1	0	1
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Power of 2	8	4	2	1
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Calculation:

$$1 \times 8$$

+

$$1 \times 4$$

+

$$0 \times 2$$

+

$$1 \times 1$$

$$8 + 4 + 0 + 1 = 13$$

Answer:

$$1101_2 = 13_{10}$$

Example 2

Convert:

101101_2

Digit	1	0	1	1	0	1
Power	32	16	8	4	2	1

Calculation:

32

+

8

+

4

+

1

Result:

45_{10}

Example 3

Convert:

11111111_2

Calculation:

128

+

64

+

32

+

16

+

8

+

4

+

2

+

1

Result:

255_{10}

6. Quick Verification

Let's verify one conversion.

Earlier:

$$25_{10} = 11001_2$$

Now convert back:

$$11001_2$$

Calculation:

$$16$$

+

$$8$$

+

$$1$$

=

$$25$$

Correct!

7. Common Mistakes

Students often:

✗ Read remainders from **top to bottom**.

Correct:

✓ Read **bottom to top**.

✗ Forget powers of 2.

Correct sequence:

1

2

4

8

16

32

64

128

✗ Add powers where the binary digit is **0**.

Remember:

Only positions with **1** contribute to the decimal value.

8. Real-Life Analogy

Imagine counting with switches.

Each switch can be:

OFF = 0

ON = 1

Example:

32 16 8 4 2 1

1 0 1 1 0 1

Only the ON switches contribute:

32

+

8

+

4

+

1

=

45

This is exactly how Binary to Decimal conversion works.



Lecture Summary

Today, we learned the first part of Number System Conversions by focusing on Decimal ↔ Binary conversions.

We studied the repeated division-by-2 method for converting Decimal to Binary and the positional-weight method for converting Binary to Decimal. We also solved several examples, discussed common mistakes, and practiced exam-style questions.

These conversion techniques are fundamental because computers process data in Binary while humans commonly use Decimal.

In the next lecture, we'll continue Number System Conversions by learning **Binary** ↔ **Octal** and **Binary** ↔ **Hexadecimal** conversions. These methods are faster than Decimal conversions and are widely used in programming, memory addressing, and debugging.

Thank you for watching, and I'll see you in the next lecture.